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Methods and apparatus of brain-like in-pixel intelligent processing system

Abstract

Aspects of present disclosure relates to a brain-like in-pixel intelligent processing system having in-pixel processing array, feature vector generator, neural network processor, and region and object of interest identifier. Each in-pixel processing unit includes photogate sensor to acquire image, average circuit to generate P element, subtraction circuit to generate F element, and absolute circuit to generate LGN element. Feature vector generator generates P, F, and LGN feature vectors from certain selected in-pixel processing units of in-pixel processing array selected according to saccadic eye movement algorithm. Neural network processor processes P, F, and LGN feature vectors and detects object, recognizes object, and determines location of object. Region and object of interest identifier identifies region and object of interest from object, and provides feedback of identified region and object of interest and processed P, F, and LGN feature vectors to the in-pixel processing array to improve the object detection and identification.

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Background/Summary

STATEMENT OF DOMESTIC PRIORITY

(1) The application claims priority to and the benefit of U.S. patent application Ser. No. 18/205,643, entitled “Methods and Apparatus of Brain-Like In-Pixel Intelligent Processing System”, filed Jun. 5, 2023 by Tuan A. Duong and QuangDao Duong, which is herein incorporated by reference in its entirety.

FIELD

(2) The present disclosure generally relates to image acquisition and processing, and more particularly to an in-pixel processing array for image acquisition and processing, object detection, classification and tracking in a brain-like in-pixel intelligent processing system built on a semiconductor chip and methods of using the brain-like in-pixel intelligent processing system.

BACKGROUND

(3) Full color image looks very satisfied via our visual systems, however, there are overwhelmed

data for a computer system to digest. The immediate price to pay includes: huge amount of data processing, large time delay, difficult to process in real time, more computer power to burn for object detection, classification and tracking, therefore, it may pose its difficulties for edge, mobile tasks.

(4) On the other hand, researches have shown that human visual systems are versatile and flexible, capable of identifying objects in a dynamic real-world environment. Rapid and accurate object recognition is achieved even though parts of the human visual system (e.g., eyes) may be fatigued, humans are unable to process detailed information at high speeds, there is limited object information to process, and/or the inability to memorize large amount of information at any given time. Saccadic eye movement is one of the bio-inspired visual models that may be emulated to extract features from captured image data based on a biological visual pathway, such as by emulating a periphery, a fovea, and a lateral geniculate nucleus of a vertebrate. The processing requirement is in the pixel level; hence it is necessary to develop a brain-like in-pixel intelligent processing system to achieve rapid and accurate object recognition by humans.

(5) Therefore, heretofore unaddressed needs still exist in the art to address the aforementioned deficiencies and inadequacies.

SUMMARY

(6) In one aspect, the present disclosure relates to an in-pixel processing array for a brain-like in-pixel intelligent processing system. In certain embodiments, the in-pixel processing array includes: a set of in-pixel processing units, a saccadic pixel column selector, and a saccadic pixel row selector for selecting an output from selected in-pixel processing units according to a saccadic eye movement algorithm based on an acquired image. The in-pixel processing array forms an image acquisition and bio-inspired processing imager configured to processes raw gray information in a pixel level. An output of the in-pixel processing array is from a set of selected in-pixel processing units of the in-pixel processing array, and the set of selected in-pixel processing units of the in-pixel processing array is selected according to the saccadic eye movement algorithm based on the acquired image.

(7) In certain embodiments, each of the in-pixel processing units includes: a photogate sensor, an average circuit, a subtraction circuit, and an absolute circuit. The photogate sensor captures a pixel of the image of an object corresponding to the in-pixel processing unit and produces an I.sub.out current of the in-pixel processing unit to the in-pixel processing array. The average circuit receives and averages I.sub.out current from photogate sensor of the in-pixel processing unit and I.sub.out current from photogate sensors of a set of neighboring in-pixel processing units around the in-pixel processing unit, and the averaged I.sub.out current from the in-pixel processing unit and the set of neighboring in-pixel processing units forms a periphery (P) element of the in-pixel processing unit. The periphery (P) element formed is used by the subtraction circuit to generate a fovea (F) element of the in-pixel processing unit. The F element generated in pixel level is sent to the absolute circuit via pixel mapping to generate lateral geniculate nucleus (LGN) element of the in-pixel processing unit.

(8) In certain embodiments, the set of in-pixel processing units of the in-pixel processing array includes: N columns, and M rows of in-pixel processing units, where N is a positive integer, and M is a positive integer. The set of neighboring in-pixel processing unit of the selected in-pixel processing units is centered at the selected in-pixel processing unit in a form of sub-window array. The set of neighboring in-pixel processing units around the in-pixel processing unit is centered at the in-pixel processing unit in a form of sub-window array. The sub-window array of the set of neighboring in-pixel processing units around the selected in-pixel processing unit includes: a circle, a square, a hexagon, and an octagon.

(9) In certain embodiments, each of the in-pixel processing units includes: a set of input channels. The set of input channels of the in-pixel processing unit includes: a first input channel from the in-pixel processing unit, a second input channel from the first neighboring in-pixel processing unit, a

third input channel from the second neighboring in-pixel processing unit, . . . , and a Q-th input channel from the (Q−1)-th neighboring in-pixel processing unit.
(10) In certain embodiments, the P element in current mode of the selected in-pixel processing unit from the average circuit is parallelly delivered in an array form through a periphery sub-window array.
(11) In certain embodiments, the F element in current mode of the selected in-pixel processing unit from the subtraction circuit is parallelly delivered in an array form through a fovea sub-window array.
(12) In one embodiment, the LGN element in current mode of the selected in-pixel processing unit from the absolute circuit is parallelly delivered in an analog array form through a LGN feature sub-window array. In another embodiment, the LGN element in current mode of the selected in-pixel processing unit from the absolute circuit is parallelly delivered off chip in a digital array form over an Analog to Digital Converter (ADC) array through the LGN feature sub-window array.
(13) In another aspect, the present disclosure relates to a brain-like in-pixel intelligent processing system. In certain embodiments, the brain-like in-pixel intelligent processing system includes: an in-pixel processing array, a feature vector generator, a neural network processor, and a region and object of interest identifier. The in-pixel processing array includes a set of in-pixel processing units, a saccadic pixel column selector, and a saccadic pixel row selector for select the output from selected in-pixel processing units according to a saccadic eye movement algorithm based on an acquired image, and each of the in-pixel processing units includes: a photogate sensor for capturing a pixel of an image of an object and produces an I.sub.out current to the in-pixel processing array, an average circuit for receiving and averaging I.sub.out current from the in-pixel processing unit and a set of neighboring in-pixel processing units around the in-pixel processing unit, and an averaged I.sub.out current forms a periphery (P) element of the in-pixel processing unit, the periphery (P) element of the in-pixel processing unit formed is used by the subtraction circuit to generate a fovea (F) element of the in-pixel processing unit, and the F element of the in-pixel processing unit generated in pixel level is sent to the absolute circuit via pixel mapping to generate lateral geniculate nucleus (LGN) element of the in-pixel processing unit. The in-pixel processing array forms an image acquisition and bio-inspired processing imager configured to processes raw gray information in a pixel level. An output of the in-pixel processing array is from a set of selected in-pixel processing units of the in-pixel processing array, and the set of selected in-pixel processing units of the in-pixel processing array is selected according to the saccadic eye movement algorithm based on the acquired image.
(14) In certain embodiments, each of the set of selected in-pixel processing units produces a periphery (P) element, a fovea (F) element, and a lateral geniculate nucleus (LGN) element of the in-pixel processing unit, the feature vector generator generates a P feature vector, an F feature vector, and an LGN feature vector of the in-pixel processing array of the image of an object from the P element, the F element, and the LGN element of each of the set of selected in-pixel processing units received, the neural network processor receives and processes the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array, detects the object, recognizes the object, and determines the location of the object, and the region and object of interest identifier provides feedback of the region and the object of interest to the in-pixel processing array to improve the object detection and identification.
(15) In certain embodiments, each selected in-pixel processing unit uses the average circuit to average the I.sub.out current from photogate sensor of the selected in-pixel processing unit and I.sub.out current from photogate sensors from a set of neighboring in-pixel processing unit around the selected in-pixel processing unit to generate a periphery (P) element of the selected in-pixel processing unit.
(16) In certain embodiments, the set of neighboring in-pixel processing unit around the selected in-pixel processing unit is centered at the selected in-pixel processing unit in a form of sub-window

array. The sub-window array of the set of neighboring in-pixel processing units around the selected in-pixel processing unit is in a circle form, a square form, a hexagon form, and an octagon form.
(17) In certain embodiments, the P element in current mode of the selected in-pixel processing unit from the average circuit is parallelly delivered in an array form through a periphery sub-window array.
(18) In certain embodiments, the F element in current mode of the selected in-pixel processing unit from the subtraction circuit is parallelly delivered in an array form through a fovea sub-window array.
(19) In one embodiment, the LGN element in current mode of the selected in-pixel processing unit from the absolute circuit is parallelly delivered in an analog array form through a LGN feature sub-window array. In another embodiment, the LGN element in current mode of the selected in-pixel processing unit from the absolute circuit is parallelly delivered off chip in a digital array form over an Analog to Digital Converter (ADC) array through the LGN feature sub-window array.
(20) In yet another aspect, the present disclosure relates to a method of using a brain-like in-pixel intelligent processing system. In certain embodiments, the method includes one of more of the following operations: exposing an object to the brain-like in-pixel intelligent processing system, the brain-like in-pixel intelligent processing system includes: an in-pixel processing array configured to acquire raw gray information of the object, and to process the raw gray information acquired in a pixel level, a feature vector generator, a neural network processor, and a region and object of interest identifier, the in-pixel processing array includes a set of in-pixel processing units and each of the in-pixel processing units includes a photogate sensor, an average circuit, a subtraction circuit, and an absolute circuit; producing, by the photogate sensor of a set of selected in-pixel processing units of the in-pixel processing array, an I.sub.out current of each selected in-pixel processing unit in response to the exposure to the object, the set of selected in-pixel processing units of the in-pixel processing array is selected according to a saccadic eye movement algorithm based on an acquired image; averaging, by the average circuit of each of the set of selected in-pixel processing units, the I.sub.out current from the selected in-pixel processing unit and a set of neighboring in-pixel processing units around the selected in-pixel processing unit to generate a periphery (P) element of each selected in-pixel processing unit; subtracting, by the subtraction circuit of each of the set of selected in-pixel processing units, the P element from the each of the set of selected in-pixel processing units to generate a corresponding fovea (F) element of each selected in-pixel processing units; producing, by the absolute circuit of each of the set of selected in-pixel processing units, a corresponding lateral geniculate nucleus (LGN) element for each of the set of selected in-pixel processing units from each of the corresponding F elements of the set of selected in-pixel processing units; generating, by the feature vector generator of the brain-like in-pixel intelligent processing system, a corresponding P feature vector, F feature vector, and LGN feature vector from each of the corresponding P elements, F elements, and LGN elements of each of the set of selected in-pixel processing units; processing, by the neural network processor, the P feature vector, the F feature vector, and the LGN feature vector of the set of selected in-pixel processing units to detect the object, identify the object, and obtain the location of the object; identifying, by the region and object of interest identifier, the region and object of interest from the processed P, F, and LGN feature vectors of the brain-like in-pixel intelligent processing system; and transmitting, by the region and object of interest identifier, the identified region and object of interest and the processed P, F, and LGN feature vectors of the brain-like in-pixel intelligent processing system back to in-pixel processing array to improve the object detection and identification.
(21) In certain embodiments, the brain-like in-pixel intelligent processing system includes: the in-pixel processing array, the feature vector generator, the neural network processor, and the region and object of interest identifier. The in-pixel processing array includes a set of in-pixel processing units, and each of the in-pixel processing units includes: the photogate sensor for capturing a pixel of an image of an object and produces an I.sub.out current to the in-pixel processing array, the

average circuit for receiving and averaging I.sub.out current from the in-pixel processing unit and a set of neighboring in-pixel processing units around the in-pixel processing unit and an averaged I.sub.out current forms a periphery (P) element, the periphery (P) element formed is used by the subtraction circuit to generate a fovea (F) element of the in-pixel processing unit; and the F element generated in pixel level is sent to the absolute circuit via pixel mapping to generate lateral geniculate nucleus (LGN) element of the in-pixel processing unit.
(22) In certain embodiments, the in-pixel processing array forms an image acquisition and bio-inspired processing imager configured to processes raw gray information in a pixel level. An output of the in-pixel processing array is from a set of selected in-pixel processing units of the in-pixel processing array, and the set of selected in-pixel processing units of the in-pixel processing array is selected according to the saccadic eye movement algorithm based on the acquired image.
(23) In certain embodiments, the feature vector generator receives and processes the P element, the F element, and the LGN element from each of selected in-pixel processing units of the in-pixel processing array, and generates the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array. The neural network processor receives and processes the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array received. The region and object of interest identifier identifies the region and object of interest from the processed P, F, and LGN feature vectors of the in-pixel processing array.
(24) In certain embodiments, each of set of selected in-pixel processing units produces a periphery (P) element, a fovea (F) element, and a lateral geniculate nucleus (LGN) element for the selected in-pixel processing unit, the feature vector generator generates a P feature vector, an F feature vector, and an LGN feature vector of the in-pixel processing array of an image of an object from the P element, the F element, and the LGN element of each of the set of selected in-pixel processing units received, the neural network processor receives and processes the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array, detects the object, recognizes the object, and determines the location of the object, and the region and object of interest identifier provides feedback of the region and the object of interest to the in-pixel processing array to improve the object detection and identification.
(25) In certain embodiments, each of the set of selected in-pixel processing unit uses the average circuit to average the I.sub.out current from the photogate sensor of the selected in-pixel processing unit and I.sub.out current from photogate sensors from a set of neighboring in-pixel processing unit around the selected in-pixel processing unit to generate a periphery (P) element of the selected in-pixel processing unit, the set of neighboring in-pixel processing unit of the selected in-pixel processing unit is centered at the selected in-pixel processing unit in a form of sub-window array, and the sub-window array of the set of neighboring in-pixel processing units of the selected in-pixel processing unit includes: a circle, a square, a hexagon, and an octagon.
(26) These and other aspects of the present disclosure will become apparent from the following description of the preferred embodiment taken in conjunction with the following drawings, although variations and modifications therein may be effected without departing from the spirit and scope of the novel concepts of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings illustrate one or more embodiments of the present disclosure, and features and benefits thereof, and together with the written description, serve to explain the principles of the present invention. Wherever possible, the same reference numbers are used throughout the drawings to refer to the same or like elements of an embodiment, and wherein:
(2) FIG. **1** illustrates an image of female face on the left and the dots and traces on the right

illustrate saccadic eye movement when human eyes look at the female face according to certain embodiments of the present disclosure;
(3) FIG. **2** shows that when looking at the female face, the human eyes only stop at these dotted locations according to certain embodiments of the present disclosure;
(4) FIG. **3** illustrates an exemplary in-pixel processing array and a saccadic pixel selection circuit having a saccadic pixel column selector and a saccadic row selector for selecting an output of the selected in-pixel processing units of the in-pixel processing array to off-chip according to certain embodiments of the present disclosure;
(5) FIG. **4** shows a block diagram of an exemplary in-pixel processing unit according to certain embodiments of the present disclosure;
(6) FIG. **5** illustrates a highlighted portion of FIG. **2** with saccadic eye movement exemplary sub-windows of neighboring in-pixel processing units of selected in-pixel processing units in different shapes according to certain embodiments of the present disclosure;
(7) FIG. **6** illustrates Saccadic Eye Movement with a group 5-pixel array on the selected in-pixel processing units P.sub.A and P.sub.B (saccadic size 5×5 array) according to certain embodiments of the present disclosure;
(8) FIG. **7** illustrates a P element and F element pixel array read-out for exemplary sub-windows array of 4×4 neighboring in-pixel processing units according to certain embodiments of the present disclosure;
(9) FIG. **8** illustrates a switching array to recorrect the P feature vector and the F feature vector order from fixed read-out mapping according to certain embodiments of the present disclosure;
(10) FIG. **9** illustrates a flow chart of output of the in-pixel processing array according to certain embodiments of the present disclosure;
(11) FIG. **10** illustrates a block diagram of a brain-like in-pixel intelligent processing system according to certain embodiments of the present disclosure; and
(12) FIG. **11** shows a flow chart of a method of using the brain-like in-pixel intelligent processing system according to certain embodiments of the present disclosure.

DETAILED DESCRIPTION

(13) The present disclosure is more particularly described in the following examples that are intended as illustrative only since numerous modifications and variations therein will be apparent to those skilled in the art. Various embodiments of the disclosure are now described in detail. Referring to the drawings, like numbers, if any, indicate like components throughout the views. As used in the description herein and throughout the claims that follow, the meaning of “a”, “an”, and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein and throughout the claims that follow, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise. Moreover, titles or subtitles may be used in the specification for the convenience of a reader, which shall have no influence on the scope of the present disclosure. Additionally, some terms used in this specification are more specifically defined below.
(14) The terms used in this specification generally have their ordinary meanings in the art, within the context of the disclosure, and in the specific context where each term is used. Certain terms that are used to describe the disclosure are discussed below, or elsewhere in the specification, to provide additional guidance to the practitioner regarding the description of the disclosure. For convenience, certain terms may be highlighted, for example using italics and/or quotation marks. The use of highlighting has no influence on the scope and meaning of a term; the scope and meaning of a term is the same, in the same context, whether or not it is highlighted. It will be appreciated that same thing can be said in more than one way. Consequently, alternative language and synonyms may be used for any one or more of the terms discussed herein, nor is any special significance to be placed upon whether or not a term is elaborated or discussed herein. Synonyms for certain terms are provided. A recital of one or more synonyms does not exclude the use of other synonyms. The use

of examples anywhere in this specification including examples of any terms discussed herein is illustrative only, and in no way limits the scope and meaning of the disclosure or of any exemplified term. Likewise, the disclosure is not limited to various embodiments given in this specification.

(15) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure pertains. In the case of conflict, the present document, including definitions will control.

(16) As used herein, “around”, “about” or “approximately” shall generally mean within 20 percent, preferably within 10 percent, and more preferably within 5 percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “around”, “about” or “approximately” can be inferred if not expressly stated.

(17) As used herein, “plurality” means two or more.

(18) As used herein, the terms “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to.

(19) As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A or B or C), using a non-exclusive logical OR. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure.

(20) As used herein, the term module may refer to, be part of, or include an Application Specific Integrated Circuit (ASIC); an electronic circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor (shared, dedicated, or group) that executes code; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip. The term module may include memory (shared, dedicated, or group) that stores code executed by the processor.

(21) The term code, as used above, may include software, firmware, and/or microcode, and may refer to programs, routines, functions, classes, and/or objects. The term shared, as used above, means that some or all code from multiple modules may be executed using a single (shared) processor. In addition, some or all code from multiple modules may be stored by a single (shared) memory. The term group, as used above, means that some or all code from a single module may be executed using a group of processors. In addition, some or all code from a single module may be stored using a group of memories.

(22) The apparatuses and methods described herein may be implemented by one or more computer programs executed by one or more processors. The computer programs include processor-executable instructions that are stored on a non-transitory tangible computer readable medium. The computer programs may also include stored data. Non-limiting examples of the non-transitory tangible computer readable medium are nonvolatile memory, magnetic storage, and optical storage.

(23) Also, it is noted that example embodiments may be described as a process depicted as a flowchart, a flow diagram, a data flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe the operations as a sequential process, many of the operations may be performed in parallel, concurrently, or simultaneously. In addition, the order of the operations may be re-arranged. A process may be terminated when its operations are completed, but may also have additional steps not included in the figure(s). A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, and the like. When a process corresponds to a function, its termination may correspond to a return of the function to the calling function and/or the main function.

(24) Example embodiments disclosed herein provide systems and methods for object recognition in image or video data. The example embodiments may be considered to be a bio-inspired model that emulates saccadic eye movements and extracts features from captured image data based on a biological visual pathway, such as by emulating the retina, fovea, and lateral geniculate nucleus

(LGN) of a vertebrate. In embodiments, an input image is provided to a saccadic eye emulator that may sequentially scan the input image to focus on one or more objects of interest and/or features within the object (e.g., when the input image includes a person, the saccadic eye emulator may focus on the eyes, nose, lips, etc., of the face of the person). Statistical data of components within the object (e.g., one or more features) may be constructed from different locations of different light intensity of the input image. In embodiments, the saccadic eye emulator generates a set of images or blocks in different locations of the input image. Principal component analysis (PCA) and/or feature extraction algorithms (FEA) may be used to obtain vertebrate features and fovea features, which are based on vertebrate retina horizontal cells and retinal ganglion cells, respectively.
(25) Example embodiments disclosed herein further provide systems and methods for identifying whole objects based on one or more (partial) object images. The bio-inspired model for extracting vertebrate features, fovea features, and LGN features is used to process one or both of the input images that form the query request and the stored images of objects in a database from which the query results will be obtained. In some embodiments, one or more searches are conducted in the stored images in the database for each image of the input images that form the query request independently of other images of the input images. And then the results of such independent searches are analyzed to determine best matches to the combination of all of the images forming the query request.
(26) The present disclosure will now be described more fully hereinafter with reference to the accompanying drawings FIGS. **1** through **12**, in which embodiments of the disclosure are shown. This disclosure may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Like numbers refer to like elements throughout.
(27) Researches have shown that human visual systems are versatile and flexible, capable of identifying objects in a dynamic real-world environment. Rapid and accurate object recognition is achieved even though parts of the human visual system (e.g., eyes) may be fatigued, humans are unable to process detailed information at high speeds, there is limited object information to process, and/or the inability to memorize large amount of information at any given time. Saccadic eye movement is one of the bio-inspired visual models that may be emulated to extract features from captured image data based on a biological visual pathway, such as by emulating a periphery, a fovea, and a lateral geniculate nucleus of a vertebrate. The processing requirement is in the pixel level; hence it is necessary to develop a brain-like in-pixel intelligent processing system to achieve rapid and accurate object recognition by humans.
(28) In certain embodiments, as shown in FIG. **1**, an image of female face is shown on the left half, and dots and traces are shown on the right half illustrate saccadic eye movement when human eyes look at the female face. Human can identify the female face based only on the much-reduced amount of data points represented by the dots and traces instead of every single pixel of the image. It appears that the dots are concentrated in feature rich area such as eyes, hair lines, nose, mouth and lips, and profile lines, as well as the spatial relationship among all these data points. It allows human to sufficiently recognize and identify the female face with such as small number of data points. Emulating the saccadic eye movement of human eyes enables a brain-like in-pixel intelligent processing system to process much less data, fast data processing and autonomous operation to facilitate the real-time feedback mechanism for adaptive learning to pave the way for machine intelligence.
(29) As shown in FIG. **2**, the data points represented by the dots are sufficient for a human to recognize and identify the female face are extracted and displayed according to certain embodiments of the present disclosure.
(30) In one aspect, as shown in FIGS. **1-9**, the present disclosure relates to an in-pixel processing array **12** for a brain-like in-pixel intelligent processing system **10**. In certain embodiments, the in-

pixel processing array **12** and the brain-like in-pixel intelligent processing system **10** is built on a semiconductor chip. In certain embodiments, the in-pixel processing array **12** includes: a set of in-pixel processing units **100**, a saccadic pixel selection circuit **1202**. The saccadic pixel selection circuit **1202** includes: a saccadic pixel column selector **12022** and a saccadic pixel row selector **12024**. The in-pixel processing array **12** forms an image acquisition and bio-inspired processing imager configured to processes raw gray information in a pixel level. An output of the in-pixel processing array **12** is from a set of selected in-pixel processing units **100** of the in-pixel processing array **12**. The set of selected in-pixel processing units **100** of the in-pixel processing array **12** is selected through the saccadic pixel column selector **12022** and the saccadic pixel row selector **12024** according to a saccadic eye movement algorithm based on an acquired image.

(31) In certain embodiments, as shown in FIG. **3**, the in-pixel processing array **12** includes has N columns and M rows of in-pixel processing units **100** P(N, M), where N and M are positive integer, P(1, 1), P(1, 2), P(1, 3), . . . , and P(N, M). The in-pixel processing array **12** forms the image acquisition and bio-inspired processing imager configured to processes raw gray information in the pixel level.

(32) In certain embodiments, as shown in FIG. **4**, each of the in-pixel processing unit **100** includes: a photogate sensor **110**, an average circuit **120**, a subtraction circuit **130**, and an absolute circuit **140**. The photogate sensor **110** captures a pixel of the image of an object corresponding to the in-pixel processing unit **100** and produces an I.sub.out current of the in-pixel processing unit **100** to the in-pixel processing array **12**. The average circuit **120** receives and averages I.sub.out current from photogate sensor of the in-pixel processing unit **100** and a set of neighboring in-pixel processing units **100** around the in-pixel processing unit **100**. The averaged I.sub.out current from the in-pixel processing unit **100** and the set of neighboring in-pixel processing units **100** forms a periphery (P) element **122** of the in-pixel processing unit **100**.

(33) In certain embodiments, as shown in FIG. **4**, the average circuit **120** receives the I.sub.out current from photogate sensor **110** of a selected in-pixel processing unit **100** itself through an input channel **1001**, and the I.sub.out current from a set of neighboring in-pixel processing units around the selected in-pixel processing unit **100** through a set of input channels **1002**, . . . , and **100**Q, where Q is a positive integer. As shown in FIG. **5**, the average circuit **120** averages the Q channels of I.sub.out current from itself as well as the group of neighboring in-pixel processing units. The averaged Q channels of lot current forms a periphery (P) element **122** and the P element **122** represents emulated saccadic eye movements of the in-pixel processing array **12**.

(34) In certain embodiments, the P element **122** of the selected in-pixel processing unit **100** generated is an input to the subtraction circuit **130** and the output of the subtraction circuit **130** forms a fovea (F) element **132** of the selected in-pixel processing unit **100**.

(35) In certain embodiments, the F element **132** of the selected in-pixel processing unit **100** in pixel level generated is sent to the absolute circuit **140** via pixel mapping to form lateral geniculate nucleus (LGN) element **142** of the selected in-pixel processing unit **100**.

(36) In certain embodiments, the image captured by the in-pixel processing array **12** includes N columns and M rows of in-pixel processing units **100**, but not all of the pixels captured is used for image processing, object recognition and identification. Only a set of selected in-pixel processing units **100** are used for image processing, object recognition and identification purposes. The set of selected in-pixel processing units **100** is selected according to a saccadic eye movement algorithm based on the acquired image by the in-pixel processing array **12** using the saccadic pixel column selector **12022** and the saccadic pixel row selector **12024** from the pixels of the acquired image.

(37) In certain embodiments, the average circuit **120** of each selected in-pixel processing unit **100** receives and averages I.sub.out current from photogate sensor of the selected in-pixel processing unit **100** and a set of neighboring in-pixel processing units **100** around the selected in-pixel processing unit **100**, as shown in FIG. **5**. As shown in FIG. **2**, the data points represented by the dots are sufficient for a human to recognize and identify the female face are extracted and

displayed according to certain embodiments of the present disclosure. The saccadic eye movement algorithm is used by the saccadic pixel column selector **12022** and the saccadic pixel row selector **12024** to select these dots as shown in FIG. **2**. FIG. **5** shows six selected in-pixel processing units: **201**, **202**, **203**, **204**, **205**, and **206**. In one embodiment, the selected in-pixel processing unit **201** is at a center of a circle **2011**, and the circle **2011** represents a set of neighboring in-pixel processing units around the selected in-pixel processing unit **201**, therefore, the neighboring in-pixel processing units sub-window array is in a round shape. In another embodiment, the selected in-pixel processing unit **202** is at a center of an octagon **2021**, and the octagon **2021** represents a set of neighboring in-pixel processing units around the selected in-pixel processing unit **202**, therefore, the neighboring in-pixel processing units sub-window array is in an octagon shape. In another embodiment, the selected in-pixel processing unit **204** is at a center of a hexagon **2041**, and the hexagon **2041** represents a set of neighboring in-pixel processing units around the selected in-pixel processing unit **204**, therefore, the neighboring in-pixel processing units sub-window array is in a hexagon shape. In yet another embodiment, each of selected in-pixel processing units **203**, **205**, and **206** is at a center of squares **2031**, **2051**, and **2061**, and the square **2031**, **2051**, and **2061** represent three sets of neighboring in-pixel processing units around selected in-pixel processing units **203**, **205**, and **206**, therefore, the neighboring in-pixel processing units sub-window arrays are in a square shape.

(38) In certain embodiments, as shown in FIG. **6**, Saccadic Eye Movement with a group 5-pixel array on the selected in-pixel processing units P.sub.A (**602**), and P.sub.B (**604**) are surrounded with a saccadic size 5×5 square sub-window arrays. The I.sub.out currents from the selected in-pixel processing units P.sub.A (**602**), is averaged with I.sub.out currents from the neighboring in-pixel processing units of a first sub-windows array **6022**. The I.sub.out currents from the selected in-pixel processing units P.sub.B (**604**), is averaged with I.sub.out currents from the neighboring in-pixel processing units of a second sub-windows array **6042**. The arrows in FIG. **6** shows the local connectivity among the neighboring in-pixel processing units **100**. This local connectivity allows the selected in-pixel processing unit P.sub.A to average the I.sub.out current from the selected in-pixel processing unit P.sub.A and the set of neighboring in-pixel processing units **100** around the selected in-pixel processing unit P.sub.A. The averaged I.sub.out current from the selected in-pixel processing unit P.sub.A and the set of neighboring in-pixel processing units **100** around the selected in-pixel processing unit P.sub.A forms the periphery (P) element **122** of the selected in-pixel processing unit P.sub.A. The periphery (P) element **122** of the selected in-pixel processing unit P.sub.A formed is used by the subtraction circuit **130** to generate a fovea (F) element **132** of the selected in-pixel processing unit P.sub.A. The F element **132** generated in pixel level is sent to the absolute circuit **140** via pixel mapping to generate lateral geniculate nucleus (LGN) element **142** of the selected in-pixel processing unit P.sub.A.

(39) In certain embodiments, as shown in FIG. **7**, a P elements and F elements Pixel array read-out is shown in FIG. **7** according to certain embodiments of the present disclosure. In FIG. **7**, the in-pixel column selector **12022** and the in-pixel row selector **12024** are used to select an m×m 2-D array (i.e. 4×4) to read-out and the same index (i,j) shares the same data bus to avoid the pixel collision. This mapping is 1-1 of any m×m pixel array and the common data bus index (i, j) is resulted in a vector array (i,j), but it may not in the right order e.g., in sub-window K, the read-out is expected: (2, 3), (2, 4), (2, 1), (2, 2), (3, 3), (3, 4), (3, 1), (3, 2), (4, 3), (4, 4), (4, 1), (4, 2), (1, 3), (1, 4), (1, 1) and (1, 2). We need to re-correct this order via a switching array in FIG. **8** and the switching array can be controlled by a finite state machine or mapping memory block.

(40) A first sub-window **1** is a 4×4 square sub-window array having 16 in-pixel processing units (1, 1), (1, 2), (1, 3), (1, 4), . . . and (4, 4), when the first four columns of the saccadic pixel column selector **12022** are switched on, and the first four rows of the saccadic pixel row selector **12024** are switched on. A K-th sub-window K is also a 4×4 square sub-window array having 16 in-pixel processing units (3, 3), (3, 4), (3, 1), (3, 2), . . . and (2, 2), when the third through sixth columns of

the saccadic pixel column selector **12022** are switched on, and the third through sixth rows of the saccadic pixel row selector **12024** are switched on. In this case, each of the first sub-window **1** through the K-th sub-window K will includes 16 in-pixel processing units: (1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), and (4, 4).
(41) In certain embodiments, a fully paralley read-out of P and F pixel array is shown in FIG. **8**, according to certain embodiments of the present disclosure. The output of the P feature vector and F feature vector **801** is controlled by an array of switches, and these switches are controlled by a finite state machine to provide the P feature vector and F feature vector in a right vector order.
(42) In certain embodiments, a flow chart of output of the in-pixel processing array **12** is shown in FIG. **9** according to certain embodiments of the present disclosure.
(43) At the block **901**, the fully parallel F feature vector in current mode generated by an array of subtraction circuit **130** of the selected in-pixel processing unit **100** is parallelly delivered to an array of absolute circuit **140**.
(44) At the block **903**, the fully parallel LGN feature vector in current mode is generated by the array of absolute circuit **140** of the selected in-pixel processing unit **100**, and the LGN feature vector in current mode is parallelly delivered as output of the in-pixel processing array **12**.
(45) There are two options for the output of the in-pixel processing array **12** off chip. The first one is analog parallel array off chip through block **905**, and the second one is digital parallel array off chip through block **911**.
(46) At block **905**, the output of the in-pixel processing array **12** from block **903** in analog current is parallelly delivered off chip through an analog array.
(47) At block **907**, the analog output of the in-pixel processing array **12** from block **903** in analog current will go through an analog to digital convertor (ADC) array to be digitized and parallelly delivered digitally to a digital array off chip of block **911**. In order to obtain accurate digital output, a maximum current value Imax from the block **903** as a reference is necessary from block **907**.
(48) In another aspect, as shown in FIG. **10**, the present disclosure relates to a brain-like in-pixel intelligent processing system **10**. In certain embodiments, the brain-like in-pixel intelligent processing system **10** includes: an in-pixel processing array **12**, a feature vector generator **14**, a neural network processor **16**, and a region and object of interest identifier **18**.
(49) In certain embodiments, the in-pixel processing array **12** includes: a set of in-pixel processing units **100**, a saccadic pixel selection circuit **1202**. The saccadic pixel selection circuit **1202** includes: a saccadic pixel column selector **12022** and a saccadic pixel row selector **12024**. The in-pixel processing array **12** forms an image acquisition and bio-inspired processing imager configured to processes raw gray information in a pixel level. An output of the in-pixel processing array **12** is from a set of selected in-pixel processing units **100** of the in-pixel processing array **12**. The set of selected in-pixel processing units **100** of the in-pixel processing array **12** is selected through the saccadic pixel column selector **12022** and the saccadic pixel row selector **12024** according to a saccadic eye movement algorithm based on an acquired image.
(50) In certain embodiments, as shown in FIG. **3**, the in-pixel processing array **12** includes has N columns and M rows of in-pixel processing units **100**, where N and M are positive integer, P(1, 1), P(1, 2), P(1, 3), . . . , and P(N, M). The in-pixel processing array **12** forms the image acquisition and bio-inspired processing imager configured to processes raw gray information in the pixel level.
(51) In certain embodiments, as shown in FIG. **4**, each of the in-pixel processing units **100** includes: a photogate sensor **110** for capturing a pixel of an image of an object and produces an I.sub.out current to the in-pixel processing array **12**, an average circuit **120** for receiving and averaging lot current from the in-pixel processing unit **100** and a set of neighboring in-pixel processing units **100** around the in-pixel processing unit **100**, and an averaged I.sub.out current forms a periphery (P) element **122** of the in-pixel processing unit **100**, the periphery (P) element **122** of the in-pixel processing unit **100** formed is used by the subtraction circuit **130** to generate a fovea (F) element **132** of the in-pixel processing unit **100**, and the F element **132** of the in-pixel

processing unit **100** generated in pixel level is sent to the absolute circuit **140** via pixel mapping to generate lateral geniculate nucleus (LGN) element **142** of the in-pixel processing unit **100**. The in-pixel processing array **12** forms an image acquisition and bio-inspired processing imager configured to processes raw gray information in a pixel level. An output of the in-pixel processing array **12** is from a set of selected in-pixel processing units **100** of the in-pixel processing array **12** as shown in FIG. **3**, and the set of selected in-pixel processing units **100** of the in-pixel processing array **12** is selected according to a saccadic eye movement algorithm based on an acquired image using the saccadic pixel column selector **12022** and the saccadic pixel row selector **12024**.

(52) The set of selected in-pixel processing units **100** has very sparse data as shown in FIG. **2**, which is reduced to 60× compared with using full color pixels. Such an innovative approach requires much less computation power for processing and less data to communicate with other systems (Bandwidth reduction).

(53) In certain embodiments, each of the set of selected in-pixel processing units **100** produces a periphery (P) element, a fovea (F) element, and a lateral geniculate nucleus (LGN) element of selected in-pixel processing unit **100**, the feature vector generator **14** generates a P feature vector, an F feature vector, and an LGN feature vector of the in-pixel processing array **12** of the image of an object from the P element, the F element, and the LGN element of each of the set of selected in-pixel processing units **100** received, the neural network processor **16** receives and processes the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array **12**, detects the object, recognizes the object, and determines the location of the object, and the region and object of interest identifier **18** provides feedback of the region and the object of interest to the in-pixel processing array **12** to improve the object detection and identification.

(54) In certain embodiments, each selected in-pixel processing unit **100** uses the average circuit **120** to average the I.sub.out current from the photogate sensor **110** of the selected in-pixel processing unit **100** and I.sub.out current from photogate sensors **110** from a set of neighboring in-pixel processing unit **100** around the selected in-pixel processing unit **100** to generate a periphery (P) element **122** of the selected in-pixel processing unit **100**.

(55) In certain embodiments, the set of neighboring in-pixel processing unit **100** around the selected in-pixel processing unit **100** is centered at the selected in-pixel processing unit **100** in a form of sub-window array, as shown in FIG. **5**. The sub-window array of the set of neighboring in-pixel processing units **100** around the selected in-pixel processing unit **100** may take the form of: (1) a circle as for the selected in-pixel processing unit **201**, (2) an octagon **2021** as for the selected in-pixel processing unit **202**, (3) an hexagon **2041** as for the selected in-pixel processing unit **204**, and (4) a square as for the selected in-pixel processing units **203**, **205**, and **206**.

(56) In certain embodiments, the P element **122** in current mode of the selected in-pixel processing unit **100** from the average circuit **120** is parallelly delivered in an array form through a periphery sub-window array.

(57) In certain embodiments, the F element **132** in current mode of the selected in-pixel processing unit **100** from the subtraction circuit **130** is parallelly delivered in an array form through a fovea sub-window array.

(58) In one embodiment, the LGN element **142** in current mode of the selected in-pixel processing unit **100** from the absolute circuit **140** is parallelly delivered in an analog array form through a LGN feature sub-window array. In another embodiment, the LGN element **142** in current mode of the selected in-pixel processing unit **100** from the absolute circuit **140** is parallelly delivered off chip in a digital array form over an Analog to Digital Converter (ADC) array through the LGN feature sub-window array.

(59) In certain embodiments, the feature vector generator **14** receives and processes the P element, the F element, and the LGN element from each of in-pixel processing units **100** of the in-pixel processing array **12** and generates the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array **12**.

(60) In certain embodiments, the neural network processor **16** receives and processes the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array **12** received according to certain brain-like intelligent processing algorithms to detect the object, to identify the object, and to determine the location of the object.
(61) In certain embodiments, the region and object of interest identifier **18** identifies the region and object of interest from the object detected and identified by the neural network processor **16** and provides feedback of the identified region and object of interest and the processed P, F, and LGN feature vectors of the brain-like in-pixel intelligent processing system to the in-pixel processing array to improve the object detection and identification.
(62) In yet another aspect, the present disclosure relates to a method of using a brain-like in-pixel intelligent processing system **10**. In certain embodiments, the brain-like in-pixel intelligent processing system **10** is built on a semiconductor chip, and the brain-like in-pixel intelligent processing system **10** includes: an in-pixel processing array **12** configured to acquire raw gray information of the object, and to process the raw gray information acquired in a pixel level, a feature vector generator **14**, a neural network processor **16**, and a region and object of interest identifier **18**. The in-pixel processing array **12** includes a set of in-pixel processing units **100**, and a saccadic pixel selection circuit **1202**. The saccadic pixel selection circuit **1202** includes: a saccadic pixel column selector **12022** and a saccadic pixel row selector **12024**. The in-pixel processing array **12** forms an image acquisition and bio-inspired processing imager configured to processes raw gray information in a pixel level. An output of the in-pixel processing array **12** is from a set of selected in-pixel processing units **100** of the in-pixel processing array **12**. The set of selected in-pixel processing units **100** of the in-pixel processing array **12** is selected through the saccadic pixel column selector **12022** and the saccadic pixel row selector **12024** according to a saccadic eye movement algorithm based on an acquired image. Each of the in-pixel processing units **100** includes a photogate sensor **110**, an average circuit **120**, a subtraction circuit **130**, and an absolute circuit **140**.
(63) In certain embodiments, the method includes one of more of the following operations: exposing an object to the brain-like in-pixel intelligent processing system **10**; producing, by the photogate sensor **110** of a set of selected in-pixel processing units **100** of the in-pixel processing array **12**, an I.sub.out current of each selected in-pixel processing unit **100** in response to the exposure to the object, the set of selected in-pixel processing units **100** of the in-pixel processing array **12** is selected according to a saccadic eye movement algorithm based on an acquired image using the saccadic pixel column selector **12022** and the saccadic pixel row selector **12024**; averaging, by the average circuit **120** of each of the set of selected in-pixel processing units **100**, the I.sub.out current from the selected in-pixel processing unit **100** and a set of neighboring in-pixel processing units **100** around the selected in-pixel processing unit **100** to generate a periphery (P) element **122** of each selected in-pixel processing unit **100**; subtracting, by the subtraction circuit **130** of each of the set of selected in-pixel processing units **100**, the P element **122** from the each of the set of selected in-pixel processing units **100** to generate a corresponding fovea (F) element **132** of each selected in-pixel processing units **100**; producing, by the absolute circuit **140** of each of the set of selected in-pixel processing units **100**, a corresponding lateral geniculate nucleus (LGN) element **142** for each of the set of selected in-pixel processing units **100** from each of the corresponding F elements **122** of the set of selected in-pixel processing units **100**; generating, by the feature vector generator **14** of the brain-like in-pixel intelligent processing system **10**, a corresponding P feature vector, F feature vector, and LGN feature vector from each of the corresponding P elements **122**, F elements **132**, and LGN elements **142** of each of the set of selected in-pixel processing units **100**; processing, by the neural network processor **16**, the P feature vector, the F feature vector, and the LGN feature vector of the set of selected in-pixel processing units **100** to detect the object, identify the object, and obtain the location of the object; identifying, by the region and object of interest identifier **18**, the region and object of interest from

the processed P, F, and LGN feature vectors of the brain-like in-pixel intelligent processing system **10**; and transmitting, by the region and object of interest identifier **18**, the identified region and object of interest and the processed P, F, and LGN feature vectors of the brain-like in-pixel intelligent processing system **10** back to in-pixel processing array **12** to improve the object detection and identification.

(64) Referring now to FIG. **11**, a flow chart of a method **1100** of using the brain-like in-pixel intelligent processing system **10** is shown according to certain embodiments of the present disclosure.

(65) At block **1102**, exposing an object to the brain-like in-pixel intelligent processing system **10**, the brain-like in-pixel intelligent processing system **10** includes an in-pixel processing array **12** having N×M in-pixel processing units **100**, and each of the in-pixel processing units **100** includes a set of input channels, a photogate sensor **110**, an average circuit **120**, a subtraction circuit **130**, and an absolute circuit **140**;

(66) At block **1104**, the saccadic pixel selection circuit **1202** of the in-pixel processing array **12** selects a group of pixels from the in-pixel processing array according to an image captured above, each pixel includes an in-pixel processing unit to capture and process the current value of the captured pixel;

(67) At block **1106**, the average circuit **120** of each in-pixel processing unit generates a periphery (p) element of the in-pixel processing unit based on received averaged I.sub.out currents, the averaged I.sub.out currents is from the I.sub.out current from the selected in-pixel processing unit and a set of neighboring in-pixel processing units around the selected in-pixel processing unit, and the averaged I.sub.out currents form an P element **122** and feed to its corresponding subtraction circuit **130** of the selected in-pixel processing units;

(68) At block **1108**, the subtraction circuit **130** of each selected in-pixel processing unit generates a fovea (F) element of the selected in-pixel processing unit based on received P elements, and feeds F elements to the absolute circuit **140** of each selected in-pixel processing unit to form lateral geniculate nucleus (LGN) elements of the selected in-pixel processing units;

(69) At block **1110**, the feature vector generator **14** is used to form P feature vectors, F feature vectors and LGN feature vectors of the selected in-pixel processing units of the in-pixel processing array **12**, based on the P elements, the F elements, and the LGN elements the set of selected in-pixel processing units of the in-pixel processing array **12**.

(70) At block **1112**, the neural network processor **16** detects the object, identifies the object, and determines the location of the object based on the P feature vector, the F feature vector, and the LGN feature vector received from the in-pixel processing array **12**; and

(71) At block **1114**, the region and object of interest identifier **18** identifies a region and object of interest from the object detected and identified by the neural network processor **16**, and provides feedback of the identified region and object of interest and the processed P, F, and LGN feature vectors of the brain-like in-pixel intelligent processing system to the in-pixel processing array **12** to improve the object detection and identification.

(72) The foregoing description of the exemplary embodiments of the disclosure has been presented only for the purposes of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Many modifications and variations are possible in light of the above teaching.

(73) The embodiments were chosen and described in order to explain the principles of the disclosure and their practical application so as to enable others skilled in the art to utilize the disclosure and various embodiments and with various modifications as are suited to the particular use contemplated. Alternative embodiments will become apparent to those skilled in the art to which the present disclosure pertains without departing from its spirit and scope. Accordingly, the scope of the present disclosure is defined by the appended claims rather than the foregoing description and the exemplary embodiments described therein.

Claims

1. An in-pixel processing array for a brain-like in-pixel intelligent processing system, comprising: a plurality of in-pixel processing units, a saccadic pixel column selector, and a saccadic pixel row selector, wherein the saccadic pixel column selector and the saccadic pixel row selector are used for selecting output from selected in-pixel processing units according to a saccadic eye movement algorithm based on an acquired image, wherein the in-pixel processing array forms an image acquisition and bio-inspired processing imager configured to processes raw gray information in a pixel level; wherein an output of the in-pixel processing array is from a plurality of selected in-pixel processing units of the in-pixel processing array, wherein the plurality of selected in-pixel processing units of the in-pixel processing array is selected according to the saccadic eye movement algorithm based on the acquired image.
2. The in-pixel processing array according to claim 1, wherein each of the in-pixel processing units comprises: a photogate sensor, wherein the photogate sensor captures a pixel of the image of an object corresponding to the in-pixel processing unit and produces an I.sub.out current of the in-pixel processing unit to the in-pixel processing array; an average circuit, wherein the average circuit receives and averages I.sub.out current from the in-pixel processing unit and a plurality of neighboring in-pixel processing units, and the averaged I.sub.out current from the in-pixel processing unit and the plurality of neighboring in-pixel processing units forms a periphery (P) element of the in-pixel processing unit; a subtraction circuit, wherein the periphery (P) element formed is used by the subtraction circuit to generate a fovea (F) element of the in-pixel processing unit; and an absolute circuit, wherein F element generated in pixel level is sent to the absolute circuit via pixel mapping to generate lateral geniculate nucleus (LGN) element of the in-pixel processing unit.
3. The in-pixel processing array according to claim 2, wherein the plurality of in-pixel processing units comprises: N columns, and M rows of in-pixel processing units, where N is a positive integer, and M is a positive integer.
4. The in-pixel processing array according to claim 3, wherein plurality of neighboring in-pixel processing units of the in-pixel processing unit is centered at the selected in-pixel processing unit in a form of sub-window array, and the sub-window array of the plurality of neighboring in-pixel processing units of the selected in-pixel processing unit comprises: a circle, a square, a hexagon, and an octagon.
5. The in-pixel processing array according to claim 4, wherein each of the in-pixel processing units comprises a plurality of input channels, and the plurality of input channels of the in-pixel processing unit comprises: a first input channel from the in-pixel processing unit, a second input channel from the first neighboring in-pixel processing unit, a third input channel from the second neighboring in-pixel processing unit, . . . , and a Q-th input channel from the (Q−1)-th neighboring in-pixel processing unit.
6. The in-pixel processing array according to claim 5, wherein the P element in current mode of the selected in-pixel processing unit from the average circuit is parallelly delivered in an array form through a periphery sub-window array.
7. The in-pixel processing array according to claim 6, wherein the F element in current mode of the selected in-pixel processing unit from the subtraction circuit is parallelly delivered in an array form through a fovea sub-window array.
8. The in-pixel processing array according to claim 7, wherein the LGN element in current mode of the selected in-pixel processing unit from the absolute circuit is parallelly delivered off chip in an analog array form through an LGN feature sub-window array.
9. The in-pixel processing array according to claim 8, wherein the LGN element in current mode of the selected in-pixel processing unit from the absolute circuit is parallelly delivered off chip in a

digital array form over an Analog to Digital Converter (ADC) array through an LGN feature sub-window array.

10. A brain-like in-pixel intelligent processing system, comprising: an in-pixel processing array, wherein the in-pixel processing array comprises a plurality of in-pixel processing units, a saccadic pixel column selector, and a saccadic pixel row selector, wherein the saccadic pixel column selector and the saccadic pixel row selector are used for selecting output from selected in-pixel processing units according to a saccadic eye movement algorithm based on an acquired image, and each of the in-pixel processing units comprises: a photogate sensor, wherein the photogate sensor captures a pixel of an image of an object and produces an I.sub.out current to the in-pixel processing array; an average circuit, wherein the average circuit receives and averages I.sub.out current from the in-pixel processing unit and a plurality of neighboring in-pixel processing units around the in-pixel processing unit and an averaged I.sub.out current forms a periphery (P) element of the in-pixel processing unit; a subtraction circuit, wherein the periphery (P) element of the in-pixel processing unit formed is used by the subtraction circuit to generate a fovea (F) element of the in-pixel processing unit; and an absolute circuit, wherein the F element of the in-pixel processing unit generated in pixel level is sent to the absolute circuit via pixel mapping to generate lateral geniculate nucleus (LGN) element of the in-pixel processing unit; and wherein the in-pixel processing array forms an image acquisition and bio-inspired processing imager configured to processes raw gray information in a pixel level; wherein an output of the in-pixel processing array is from a plurality of selected in-pixel processing units of the in-pixel processing array, and the plurality of selected in-pixel processing units of the in-pixel processing array is selected according to the saccadic eye movement algorithm based on the acquired image; a feature vector generator, wherein the feature vector generator receives and processes the P element, the F element, and the LGN element from each of selected in-pixel processing units of the in-pixel processing array, and generates the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array; a neural network processor, wherein the neural network processor receives and processes the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array received; a region and object of interest identifier, wherein the region and object of interest identifier identifies the region and object of interest from the processed P, F, and LGN feature vectors of the in-pixel processing array, and wherein each of the plurality of selected in-pixel processing units produces a periphery (P) element, a fovea (F) element, and a lateral geniculate nucleus (LGN) element of the in-pixel processing unit, the feature vector generator generates a P feature vector, an F feature vector, and an LGN feature vector of the in-pixel processing array of the image of the object from the P element, the F element, and the LGN element of each of the plurality of selected in-pixel processing units received, the neural network processor receives and processes the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array, detects the object, recognizes the object, and determines the location of the object, and the region and object of interest identifier provides feedback of the region and the object of interest to the in-pixel processing array to improve the object detection and identification.

11. The brain-like in-pixel intelligent processing system according to claim 10, wherein each of the selected in-pixel processing unit uses the average circuit to average the I.sub.out current from the photogate sensor of the selected in-pixel processing unit and lot current from photogate sensors from a plurality of neighboring in-pixel processing unit around the selected in-pixel processing unit to generate a periphery (P) element of the selected in-pixel processing unit, the plurality of neighboring in-pixel processing unit around the selected in-pixel processing unit is centered at the selected in-pixel processing unit in a form of sub-window array, and the sub-window array of the plurality of neighboring in-pixel processing units around the selected in-pixel processing unit comprises: a circle, a square, a hexagon, and an octagon.

12. The brain-like in-pixel intelligent processing system according to claim 11, wherein the P

element in current mode of the selected in-pixel processing unit from the average circuit is parallelly delivered in an array form through a periphery sub-window array.
13. The brain-like in-pixel intelligent processing system according to claim 12, wherein the F element in current mode of the selected in-pixel processing unit from the subtraction circuit is parallelly delivered in an array form through a fovea sub-window array.
14. The brain-like in-pixel intelligent processing system according to claim 13, wherein the LGN element in current mode of the selected in-pixel processing unit from the absolute circuit is parallelly delivered off chip in an analog array form through an LGN feature sub-window array.
15. The brain-like in-pixel intelligent processing system according to claim 13, wherein the LGN element in current mode of the selected in-pixel processing unit from the absolute circuit is parallelly delivered off chip in a digital array form over an Analog to Digital Converter (ADC) array through an LGN feature sub-window array.
16. A method of using a brain-like in-pixel intelligent processing system, comprising: exposing an object to the brain-like in-pixel intelligent processing system, wherein the brain-like in-pixel intelligent processing system comprises: an in-pixel processing array configured to acquire raw gray information of the object and to process the raw gray information acquired in a pixel level, a feature vector generator, a neural network processor, and a region and object of interest identifier, wherein the in-pixel processing array comprises a plurality of in-pixel processing units, a saccadic pixel column selector, and a saccadic pixel row selector for selecting output from selected in-pixel processing units according to a saccadic eye movement algorithm based on an acquired image, and each of the plurality of in-pixel processing units comprises a photogate sensor, an average circuit, a subtraction circuit, and an absolute circuit; producing, by the photogate sensor of a plurality of selected in-pixel processing units, an I.sub.out current of each selected in-pixel processing unit in response to the exposure to the object, wherein the plurality of selected in-pixel processing units of the in-pixel processing array is selected according to the saccadic eye movement algorithm based on the acquired image; averaging, by the average circuit of each of the plurality of selected in-pixel processing units, the I.sub.out current from the selected in-pixel processing unit and a plurality of neighboring in-pixel processing units around the selected in-pixel processing unit to generate a periphery (P) element of each selected in-pixel processing unit; subtracting, by the subtraction circuit of each of the plurality of selected in-pixel processing units, the P element from the each of the plurality of selected in-pixel processing units to generate a corresponding fovea (F) element for each of the plurality of selected in-pixel processing units; producing, by the absolute circuit of each of the plurality of selected in-pixel processing units, a corresponding lateral geniculate nucleus (LGN) element for each of the plurality of selected in-pixel processing units from each of the corresponding F elements of the plurality of selected in-pixel processing units; generating, by the feature vector generator of the brain-like in-pixel intelligent processing system, a corresponding P feature vector, F feature vector, and LGN feature vector from each of the corresponding P elements, F elements, and LGN elements of each of the plurality of selected in-pixel processing units; processing, by the neural network processor, the P feature vector, the F feature vector, and the LGN feature vector of the plurality of selected in-pixel processing units to detect the object, to identify the object, and obtain the location of the object; identifying, by the region and object of interest identifier, the region and object of interest from the processed P, F, and LGN feature vectors of the brain-like in-pixel intelligent processing system; and transmitting, by the region and object of interest identifier, the identified region and object of interest and the processed P, F, and LGN feature vectors of the brain-like in-pixel intelligent processing system back to in-pixel processing array to improve the object detection and identification.
17. The method according to claim 16, wherein the brain-like in-pixel intelligent processing system comprises: the in-pixel processing array, wherein the in-pixel processing array comprises a plurality of in-pixel processing units, a saccadic pixel column selector, and a saccadic pixel row selector, wherein the saccadic pixel column selector and the saccadic pixel row selector are used

for selecting output from selected in-pixel processing units according to the saccadic eye movement algorithm based on the acquired image, and each of the in-pixel processing units comprises: the photogate sensor, wherein the photogate sensor captures a pixel of an image of an object and produces an I.sub.out current to the in-pixel processing array; the average circuit, wherein the average circuit receives and averages I.sub.out current from the in-pixel processing unit and a plurality of neighboring in-pixel processing units around the in-pixel processing unit and an averaged I.sub.out current forms a periphery (P) element; the subtraction circuit, wherein the periphery (P) element generated is used by the subtraction circuit to generate a fovea (F) element of the in-pixel processing unit; and the absolute circuit, wherein the F element generated in pixel level is sent to the absolute circuit via pixel mapping to generate lateral geniculate nucleus (LGN) element of the in-pixel processing unit; and wherein the in-pixel processing array forms an image acquisition and bio-inspired processing imager configured to processes raw gray information in a pixel level; wherein an output of the in-pixel processing array is from a plurality of selected in-pixel processing units of the in-pixel processing array, and the plurality of selected in-pixel processing units of the in-pixel processing array is selected according to the saccadic eye movement algorithm based on the acquired image; the feature vector generator, wherein the feature vector generator receives and processes the P element, the F element, and the LGN element from each of selected in-pixel processing units of the in-pixel processing array, and generates the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array; the neural network processor, wherein the neural network processor receives and processes the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array received; the region and object of interest identifier, wherein the region and object of interest identifier identifies the region and object of interest from the processed P, F, and LGN feature vectors of the in-pixel processing array, and wherein each of the plurality of selected in-pixel processing units produces a periphery (P) element, a fovea (F) element, and a lateral geniculate nucleus (LGN) element for the selected in-pixel processing unit, the feature vector generator generates a P feature vector, an F feature vector, and an LGN feature vector of the in-pixel processing array of the image of the object from the P element, the F element, and the LGN element of each of the plurality of selected in-pixel processing units received, the neural network processor receives and processes the P feature vector, the F feature vector, and the LGN feature vector of the in-pixel processing array, detects the object, recognizes the object, and determines the location of the object, and the region and object of interest identifier provides feedback of the region and the object of interest to the in-pixel processing array to improve the object detection and identification.

18. The method according to claim 17, wherein each of the selected in-pixel processing unit uses the average circuit to average the I.sub.out current from the photogate sensor of the selected in-pixel processing unit and I.sub.out current from photogate sensors from a plurality of neighboring in-pixel processing unit of the selected in-pixel processing unit to generate a periphery (P) element of the selected in-pixel processing unit.

19. The method according to claim 18, wherein plurality of neighboring in-pixel processing unit around the selected in-pixel processing unit is centered at the selected in-pixel processing unit in a form of sub-window array.

20. The method according to claim 19, wherein the sub-window array of the plurality of neighboring in-pixel processing units around the selected in-pixel processing unit comprises: a circle, a square, a hexagon, and an octagon.